

SIT and P versus NP as a Coherence-Guided Computation Program

A proof-obligation, resource-accounting, and adverse-benchmark agenda

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Abstract

A proposed answer can sometimes be checked much faster than it can be found. P versus NP asks whether that apparent difference is fundamental in the standard model of efficient computation. Super Information Theory (SIT) motivates a physical interpretation of the difference: checking tests a supplied pattern, whereas finding must organize a constrained system into a pattern that can be read as a witness. This paper turns that intuition into a disciplined research program rather than announcing a solution.

The paper first states the conventional complexity-theory target. It then defines a minimal Boolean proxy in which unsatisfied constraints create mismatch, improving moves reduce mismatch, and agreement among improving moves provides a bounded local coherence diagnostic. Exhaustive tests on four tiny formula families are intentionally adverse. One-bit mismatch descent reaches a witness from every initial state of a friendly formula, but reaches the unique witness from only 21 of 64 initial states in a six-variable implication chain and 13 of 32 states in a fixed five-variable 3-CNF. The same cases contain 16 and 6 non-witness local minima, respectively; an unsatisfiable formula shows that a minimum need not encode a witness. These finite results do not establish an asymptotic lower bound. They show why relaxation toward lower mismatch is not, by itself, an algorithm for NP search.

The constructive proposal is therefore conditional. A coherence-guided route could have a standard complexity consequence only after it supplies a uniform polynomial-time encoder, a fully specified evolution or polynomial-size macro certificate, polynomial precision and robustness margins, a constructive readout, ordinary witness verification, and an explicit account of relativization, natural-proofs, algebrization, mixing, locality, and counting barriers. The strongest formulation, called Path B, asks for certified macro readout without requiring a verifier to replay the full microdynamic history. All Path B obligations and all seven associated kernel obligations remain open. Two replayed Lean artifacts are retained as an example of correct scope: one refutes a finite Farey bridge and the other proves a narrower finite repair, with no P-versus-NP consequence. The contribution is a falsifiable, resource-accounted program for turning a physical intuition into claims that could eventually be judged in ordinary complexity theory.

computation; physical computation; certified readout; analog complexity; constraint satisfaction; adverse benchmark; proof obligations

Keywords: P versus NP; Super Information Theory; coherence-guided

Scope and evidence boundary

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1. The ordinary problem: finding a key and checking a key

Imagine a lock with many tumblers. If someone gives you a candidate key, you can try it and quickly learn whether it opens the lock. Without the key, you may have to examine an enormous number of possibilities. The quick test and the difficult search are different operations.

Computer science makes that distinction precise. A *decision problem* asks for a yes-or-no answer. For a yes-instance, a *witness* is a finite object that lets a verifier confirm the answer. If x is an instance, y is a witness, V is a deterministic verifier, and p is a polynomial, then the defining witness form used in this paper is

$$x \in L \iff \exists y (|y| \leq p(|x|) \wedge V(x, y) = 1). \quad (1)$$

The class P contains decision problems solvable by a deterministic Turing-machine algorithm in polynomial time. The class NP contains decision problems whose yes-instances have polynomial-size witnesses checkable in polynomial time. P is contained in NP. Whether P equals NP remains open [1,2].

Cook and Karp established the central role of NP-complete problems [3,4]. If one NP-complete problem has a deterministic polynomial-time algorithm, then every problem in NP does. Conversely, evidence that a physical device often finds good answers, rapidly lowers an energy, or solves a restricted family does not establish such an algorithm. A standard result must include the cost of constructing the instance, representing every coefficient, running or certifying the process, extracting the answer, and checking it.

This paper does not prove P equals NP, prove P differs from NP, or replace the Turing-machine question with a different physical question. Its aim is to ask what a particular physical intuition would have to become before it could address the conventional problem.

2. The computational problem behind the physical intuition

A constraint-satisfaction problem can be pictured as a landscape. Some configurations violate many constraints; some violate few; a satisfying configuration violates none. It is tempting to say that a physical system will simply settle toward the lowest mismatch and thereby find an answer. That temptation hides several problems.

First, a landscape can have local minima. A configuration can be worse than a true solution but admit no improving local move. Second, a globally lowest state may not encode a witness if the instance is unsatisfiable or the encoding is incorrect. Third, even when the physics reaches a useful state, the required time, precision, energy, preparation, and readout can grow exponentially. Fourth, the landscape itself may take exponential work or nonuniform advice to construct. Fifth, a compact-looking observable can conceal a hard counting problem.

These are not objections to physical computation. They are the accounting rules that make a physical-computation claim meaningful. The theory of computation over the reals illustrates the point: the Blum-Shub-Smale model has an explicitly different machine and cost structure [8]. Conversely, Bournez, Graça, and Pouly recover ordinary polynomial time for a class of polynomial ordinary differential equations only after specifying a robust trajectory length as the relevant resource [9]. A continuous process becomes a complexity-theoretic object through definitions and bounds, not through continuity alone.

3. Established accounts before SIT terminology

Several established research programs already connect computation to landscapes, dynamics, and physical resources:

- Boolean satisfiability supplies a canonical witness-finding problem.
- Ising and related encodings can represent many optimization problems, but representation does not guarantee efficient equilibration or readout [10,11].
- Markov-chain methods can yield polynomial-time approximation in specified regimes, while nearby regimes can be computationally hard [12,13].
- Spin-glass and random-constraint models exhibit clustering, frozen variables, metastability, and phase changes that can frustrate local search [14].
- Complexity barriers show that broad classes of proof strategies are insufficient without additional structure: relativization [5], natural proofs [6], and algebrization [7].
- Counting and partition-function observables require their own analysis. Exact counting can be #P-complete, while some model-specific approximation problems admit efficient randomized algorithms [12,15].

The lesson is not that all physical search must fail. It is that the model, regime, update rule, resource measure, and readout determine the claim.

4. The proposed operation in plain language

The proposal begins with six ordinary operations.

1. Translate the input constraints into a finite, checkable model.
2. Represent how badly a candidate state violates those constraints.
3. Identify changes that reduce the mismatch.
4. Allow compatible changes to reinforce one another rather than treating every candidate move as an isolated guess.
5. Continue until a stable macroscopic pattern can be read.
6. Submit the extracted candidate to the ordinary verifier.

The central hypothesis is that useful structure may appear when many local recommendations agree and can be combined into a coordinated change. A structured instance might then expose a strong, compressible direction of improvement. A glassy instance might instead present mutually incompatible recommendations, narrow bottlenecks, or a readout whose apparent simplicity conceals a large computational cost.

Nothing in that description yet says that the system is fast, uniform, robust, or correct. It only identifies a possible operation: constrained mismatch is transformed into coordinated change, and coordinated change is transformed into a candidate witness.

5. Naming the proposal: coherence-guided computation

This paper calls the operation **coherence-guided computation** (CGC). In the broader SIT vocabulary, an encoded instance supplies an initial mismatch field Φ_0 , a time-dependent usable-coherence field R_{coh} , and, in mature physical models, potentially a time-density field ρ_t . “Coherence” here does not mean that similarity automatically solves a problem. It names the degree to which candidate updates can be coordinated without destroying the constraints already satisfied.

5.1 A bounded Boolean proxy

To prevent the physical vocabulary from outrunning its content, consider a minimal finite proxy. Let a CNF formula F have clauses $C_1, \dots, C_m$ and Boolean assignment s . Its mismatch energy is the number of unsatisfied clauses:

$$E_F(s) = \sum_{j=1}^m \mathbf{1}[C_j(s) = 0]. \quad (2)$$

For variable i , let $s^{\oplus i}$ be s with bit i flipped. The one-bit improvement is

$$\Delta_i(s) = E_F(s) - E_F(s^{\oplus i}). \quad (3)$$

A deliberately modest local recommendation-coherence score is

$$R_{\text{loc}}(s) = \begin{cases} \frac{\max_i \Delta_i^+(s)}{\sum_i \Delta_i^+(s)}, & \sum_i \Delta_i^+(s) > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

$$\Delta_i^+ = \max(\Delta_i, 0).$$

The score is high when the improving recommendations concentrate on a small set of moves. It is zero when no one-bit move improves the state. It is only a finite diagnostic; it is not identified with a measured physical R_{coh} .

The CGC-0 update flips the lowest-index variable among the maximum-gain strictly improving moves:

$$s_{t+1} = \begin{cases} s_t^{\oplus i^*}, & i^* = \min \arg \max_i \Delta_i(s_t), \max_i \Delta_i(s_t) > 0, \\ s_t, & \max_i \Delta_i(s_t) \leq 0. \end{cases} \quad (5)$$

This update is intentionally weak and reproducible. It tests the proposition that lowering mismatch is sufficient. It does not implement the full SIT proposal, nonlocal moves, noise, annealing, tunneling, parallel analog evolution, or a macro certificate.

A useful landscape diagnostic is the smallest peak mismatch on any one-bit path from s to the witness set W_F :

$$B_F(s) = \min_{\gamma: s \rightsquigarrow W_F} \max_{u \in \gamma} E_F(u), \quad (6)$$

$$W_F = \{u : E_F(u) = 0\}.$$

When every route to a witness must first increase mismatch, strictly downhill CGC-0 cannot take that route. When W_F is empty, the barrier is undefined and no minimum can be accepted as a witness.

5.2 What the richer proposal would add

The richer CGC program asks whether a physically specified system can combine many compatible local differentials into a collective update, preserve useful global information, and generate a readout certificate cheaper than microstate replay. Candidate additions include nonlocal but polynomially-described couplings, a dissipative term, a U(1)-like phase variable, a defined invariant or slow manifold, and a finite set of macro variables. Each addition is a resource and must be represented in the model and its cost.

This is where the SIT terms earn their role. A phase-wave differential can serve as one candidate representation of receiver-relative mismatch; a coherence current can describe coordinated propagation of an update; and a time-density field can parameterize a proposed physical timing geometry. None of those names grants computational power. The mathematical model must say how they are encoded, evolved, measured, and certified.

6. Bounded adverse benchmark

The accompanying program exhausts every assignment of four fixed formulas, applies Equation (5), and records witnesses, non-witness one-flip local minima, success basins, step counts, and the minimax quantity in Equation (6). The normal and optimized Python runs produce byte-identical reports. Seven tests pass in each mode.

Formula	Variables	Clauses	States	Witnesses	Non-witness local minima	Starts reaching a witness	Maximum CGC-0 steps
Friendly one-clause 3-SAT	3	1	8	7	0	8/8	1
Implication chain	6	6	64	1	16	21/64	3
Fixed greedy-trap 3-CNF	5	20	32	1	6	13/32	3
Contradictory unit clauses	1	2	2	0	2	0/2	0

The friendly formula behaves as intuition suggests. The implication chain is more revealing. It has a unique all-true witness, yet a strictly improving one-bit rule stops at 16 distinct non-witness minima and succeeds from fewer than one third of its states. The fixed five-variable 3-CNF likewise has a unique witness but six non-witness local minima. The contradictory formula has no witness although both assignments minimize its mismatch.

These results establish a narrow adverse fact:

A deterministic rule that repeatedly lowers the number of unsatisfied clauses does not reliably find a witness, even in the bounded families tested here.

They do **not** establish that all local algorithms fail, that all analog systems mix slowly, that every richer coherence operation is reducible to one-bit descent, or that P differs from NP. Their constructive purpose is to force the next proposal to identify the extra operation. If collective coherence crosses a barrier, what coupling performs the crossing? How is that coupling constructed? If a macro certificate avoids the crossing, how is the certificate produced and checked? If noise or tunneling helps, what success probability, time, precision, and repetition cost does it require?

7. The full resource ledger

Wall-clock time alone is not a sufficient cost measure for a physical solver. For an encoded instance x , define a resource vector

$$\mathcal{R}(x) = (b_{\text{enc}}, T, \Pi, \mathcal{E}, \mathcal{V}, b_{\text{read}}, p_{\text{succ}}), \quad (7)$$

where b_{enc} is the bit length and construction cost of the model, T is evolution or certificate depth, Π is required precision, $\mathcal{E}$ is energy / action, $\mathcal{V}$ is physical volume or parallel hardware, b_{read} is readout and postprocessing cost, and p_{succ} is success probability. A conventional polynomial-time claim needs polynomial bounds on every resource that can hide computation, plus a polynomially bounded amplification cost when success is probabilistic.

The earlier SIT development proposed a physical quantity called **coherence-time action**. A disciplined version is

$$\begin{aligned} C_{\text{SIT}}(x) &= \inf_{\gamma \in \Gamma_x} \int_0^{T_\gamma} \mathcal{A}_{\text{coh}}(\gamma(t)) dt, \\ C_{\text{SIT}}(n) &= \max_{|x|=n} C_{\text{SIT}}(x), \end{aligned} \quad (8)$$

where Γ_x is the declared class of admissible successful trajectories and $\mathcal{A}_{\text{coh}}$ is a model-defined nonnegative action density. Equation (8) is a proposed engineering / physics complexity measure, not a known complexity class. Its scientific value depends on operationally defining Γ_x , $\mathcal{A}_{\text{coh}}$, success, and every hidden preparation and readout cost.

The source genealogy also proposed a schematic convergence relation

$$T_\varepsilon(x) \leq \kappa \frac{C_P(\Omega_x)}{\mathcal{I}(x)} \log \left(\frac{E_{\text{initial}} - E_\star}{\varepsilon} \right). \quad (9)$$

Here $C_P(\Omega_x)$ is a normalized domain/Poincaré factor, $\mathcal{I}(x)$ is a positive information-curvature, contraction, or spectral-gap quantity, $E_\star$ is the target energy, and κ is a model constant. Equation (9) is **not proved in this paper**. It is retained because it says what a useful theorem would have to control: geometry, contraction, target accuracy, and worst-case scaling. The quantities must be defined with consistent dimensions or nondimensionalization before the inequality can be tested. A useful-instance correlation is not a worst-case bound, and an exponentially small $\mathcal{I}(x)$ would erase the hoped-for speedup.

An earlier diagnostic branch factorized an instance-level coherence index as $C(I) = g A M$: a coupling or spectral factor g , an alignment factor A , and a memory / embedding match factor M . That factorization is retained as a measurement proposal, not a derived invariant. Each factor must be defined separately for Boolean, neural, machine-learning, or physical systems; the product must outperform simpler statistics such as spectral gap, clause density, condition number, and baseline solver features on held-out instances. Until then, $C(I)$ and $\mathcal{I}(x)$ are candidate coordinates for practical instance difficulty, not complexity-class parameters.

8. Path B: certified macro readout

Replaying every microscopic degree of freedom may be unnecessary if a short certificate proves the relevant macroscopic history and readout. This is the motivation for **Path B**. Its target contract is

$$\begin{aligned}
M_x &= \text{Enc}(V, x), \\
\text{Cert}(M_x, c) &= 1, \\
\text{Read}(M_x, c) &= y, \\
V(x, y) &= 1.
\end{aligned} \tag{10}$$

To have the desired consequence, Enc, Cert, and Read must be uniform deterministic polynomial-time procedures; M_x , c , and every coefficient must have polynomial bit length; and the contract must hold for arbitrary yes-instances of a standard NP-complete witness problem. A certificate that can be checked quickly but cannot be *found* quickly does not by itself solve the search problem. Path B therefore requires either a polynomial-time certificate constructor or a physical process whose complete resource vector in Equation (7) is polynomially bounded and whose output is the certificate.

Let S be a full state and $K(S)$ a finite macro projection. Exact closure would take the form

$$K(\text{Evo}_t(S)) = \widehat{\text{Evo}}_t(K(S)), \tag{11}$$

or a rigorously bounded approximate version with an error budget small enough to preserve the final witness. Without Equation (11), macro evolution may silently require the full microhistory.

Robust readout also needs a separation margin. If $\mathcal{K}_y$ is the macro region decoded as witness y , a representative requirement is

$$\text{dist}\left(\mathcal{K}_y, \bigcup_{z \neq y} \mathcal{K}_z\right) \geq \frac{1}{q(|x|)} \quad \text{for some polynomial } q. \tag{12}$$

Other margin formulations are possible, but they must prevent exponentially precise distinctions from being treated as free.

8.1 The information-throughput branch

The source program also explored a model-relative counting lemma. Let $\chi_M(f)$ be the smallest number of cells in an M -definable partition of the input space such that f is constant on every cell. If one step of an M -algorithm yields at most b_M bits of transcript information, then a deterministic decision tree obeys

$$\text{time}_M(f) \geq \left\lceil \frac{\log_2 \chi_M(f)}{b_M} \right\rceil. \tag{13}$$

Equation (13) is a finite counting statement once M , its admissible tests, and b_M are fixed. It recovers familiar partition or decision-tree bounds in appropriate restricted models. It does not separate P from NP: reaching that target would require a model that faithfully simulates all of P, a superpolynomial lower bound for an NP-complete language, and a proof that the argument escapes the known barriers. The branch is nevertheless useful because it turns “information throughput” into an explicit resource rather than a metaphor.

The Path B theorem target can now be stated without ambiguity:

If arbitrary NP-verifier instances admit the uniform encoding, polynomial resource bounds, macro closure or polynomial-size certificate construction, robust readout, and ordinary verification described above, then the route would yield a conventional polynomial-time witness-finding procedure.

This is a conditional target. No clause of the target is claimed complete.

9. Kernel obligations K0-K6

The internal “Kernel Theorem” language is retained as a seven-part certificate target. It is not the name of a completed theorem.

ID	Obligation	Completion criterion	Status
K0	Polynomial description	The action, coefficients, initial state, couplings, and macro variables are constructed uniformly with polynomial bit length.	Open
K1	Macro closure	Equation (11), or a sufficient bounded approximation, is proved without microstate replay.	Open
K2	Coercive witness minima	Valid witnesses are stable targets and non-witness states are separated by a polynomially robust criterion.	Open
K3	Uniform macro gap	Convergence, contraction, or certificate depth has a worst-case polynomial bound on adversarial families.	Open
K4	Macro-to-micro sandwich	The macro certificate faithfully controls every microquantity needed for correctness.	Open
K5	Constructive readout	Counts, holonomies, integrals, or other invariants used by Read are polynomially computable and robust.	Open
K6	Nonrelativizing structure	The route identifies the exact syntactic, arithmetic, geometric, or topological operation unavailable to a black-box oracle method.	Open

“Kernel” is useful only if it compresses the dynamics without discarding the information needed to prove correctness. A projection that merely names a small set of variables is not yet a computational reduction.

10. Barrier and repair matrix

The matrix below is part of the proposed method, not a defensive appendix. Each row states how an apparently successful physical story can fail.

Barrier	Conventional anchor	Failure mode	Required repair
Relativization	Baker-Gill-Solovay [5]	A black-box clause-query argument survives in oracle worlds with conflicting P/NP behavior.	Exhibit a precise nonrelativizing operation.
Natural proofs	Razborov-Rudich [6]	A broad constructive largeness property cannot support the intended circuit lower bound under the stated pseudorandomness assumptions.	Show why the property is outside the barrier or make no separation claim.
Algebrization	Aaronson-Wigderson [7]	An algebraic reformulation still belongs to a method class known to be insufficient.	Identify and prove the non-algebrizing step.
Uniformity	Standard machine accounting [2,4]	The model family hides advice or exponential setup in hand-built couplings.	Give a polynomial-time encoder and coefficient generator.
Locality and gadget blowup	Reduction theory [4,10]	Local gadgets enlarge the system, introduce metastability, or require exponentially tuned constants.	Bound size, degree, coefficient bits, and induced dynamics.
Finite precision	Real / ODE computation [8,9]	Exact real constants or unstable initial conditions conceal exponential information.	Define representation, perturbation tolerance, and polynomial margins.
Torpid mixing	Markov-chain and spin-system results [12,13]	Local relaxation crosses a bottleneck with exponentially small probability or gap.	Prove a polynomial gap, contraction, or certificate alternative on the encoded family.
Energy barriers and glassiness	Spin-glass / CSP literature [11, 14]	Descent halts in metastable non-witness states, as CGC-0 already does in tiny cases.	Specify collective moves and account for their construction and execution.
Readout / counting hardness	Valiant; Jerrum-Sinclair [12, 15]	A macro observable hides exact or approximate counting work.	Define the observable, approximation guarantee, and polynomial certificate.
Degeneracy and near-collisions	Robust-computation requirement	Witness and non-witness basins are closer than polynomial precision can resolve.	Prove a margin such as Equation (12).
Unsatisfiable instances	SAT baseline [3,4]	Relaxation finds a minimum but no witness exists.	Include a sound decision certificate or restrict and declare the task.
Probability amplification	Randomized-algorithm accounting	A tiny success probability is treated as a fast solver.	Bound repetitions, randomness, and verification cost.

No row is closed by analogy. A future proof artifact must identify the exact model and theorem that closes it.

11. Discriminating tests and failure conditions

The proposal becomes scientifically useful by risking failure. The next computational tier should implement at least three update families on the same encoded instances:

1. CGC-0 deterministic local descent;

2. a declared collective-update rule with polynomial construction cost;
3. a conventional baseline such as WalkSAT, simulated annealing, or an appropriately chosen message-passing method.

The test set should include:

- random 3-SAT across easy, clustered, and frozen regimes;
- planted locked constraint-satisfaction problems;
- antiferromagnetic two-spin and hard-core instances in precisely stated graph and parameter regimes;
- adversarial verifier encodings designed to expose gadget blowup;
- satisfiable and unsatisfiable paired instances;
- instances with controlled near-degenerate readouts.

For each family, report the full vector in Equation (7), not only elapsed time. Estimate scaling with confidence intervals; disclose timeouts and failed starts; verify every candidate by the ordinary verifier; and distinguish heuristic instance performance from worst-case proof.

The CGC interpretation is weakened or falsified in a stated model if:

- its coherence diagnostic does not predict performance better than simpler landscape statistics out of sample;
- the collective rule offers no reproducible advantage over matched baselines;
- coefficient precision, physical volume, preparation, or readout grows superpolynomially;
- the macro variables fail closure or lose witness-relevant information;
- the proposed separation margin becomes exponentially small;
- a claimed uniform gap collapses on an adversarial family;
- the certificate cannot be constructed or checked in polynomial resources.

A bounded empirical advantage would still be valuable as an engineering result. It should not be promoted to P equals NP .

12. A formal scope case study: obstruction, then narrow repair

This release retains two Lean-checked finite Farey artifacts because they illustrate the paper’s methodological rule.

The first artifact, BC-054IS, supplies the concrete endpoints $N = 3$, $a = (0, 1)$, and $b = (2, 3)$ and refutes one proposed arbitrary-endpoint shape bridge. The second, BC-055O, proves a left-endpoint bounded inverse, a direct certified-mediator constructor premise, and a listed-mediator target in the current finite model. Both exact source files replay under Lean 4.30.0 with pinned Mathlib; theorem-level axiom reports complete.

The repair does not reverse the obstruction. It uses a different and narrower Euclidean route. Neither artifact provides an NP encoder, CGC evolution, certificate constructor, witness readout, K0-K6 proof, P-versus-NP theorem, RH result, Hilbert-Pólya result, or full-SIT theorem.

This is the intended proof culture: a failed bridge remains failed; a narrower result remains available at its exact scope; and higher-level language does not inherit a theorem that was never proved.

12.1 Relationship to the wider SIT proof program

The larger proof ledgers are used here for scope discipline, not as complexity-theory evidence.

Wider-program status	Consequence for this paper
SIT79 is fixed as a source-backed specified model surface in the capped ledger.	It supplies source stability, not an NP encoder or solver.
Abstract finite projection kernels are accepted only under their exact hypotheses.	They motivate conditional theorem discipline but do not establish K0-K6 here.
CT-BOS, direct finite v75, CHT, and private finite HSIT routes are retired, capped, or assumption-only under the current evidence.	Their names cannot be imported as hidden proof steps.
The Coherence Tower remains unproved except by caps below full SIT.	It supplies no uniform macro gap for Path B.
Physical SIT is separated from the mathematical theorem package.	A physical search story cannot automatically become a Turing-model result.
SIT / Riemann, RH, Hilbert-Pólya, determinant/Xi identity, and publication clearance are not promoted by the current ledgers.	No such result is evidence for P versus NP.

12.2 Historical branches retained without promotion

The private development archive contains two stronger directions that are important to preserve even though neither is adopted as a result.

One branch proposed a global “finite napkin” constraint, $Q_* = \int_{\Sigma} \rho_t \sqrt{g^{(3)}} d^3x = \text{constant}$, and used it to motivate a **SIT-Complexity Hypothesis**: generic NP-complete search might require superpolynomial coherence-time action under local conserved dynamics and a fixed global budget. The archive itself labels the constraint an added axiom rather than a consequence already derived from SIT. No reduction, lower-bound theorem, or empirical scaling result establishes the hypothesis.

The other branch proposed the opposite possibility: rapid nonlocal or “entropic tunneling” dynamics might make encoded NP search polynomial. The same source exchange identifies the missing uniform encoding, finite precision, convergence, and readout proofs. The present paper does not erase either intuition. It translates their disagreement into the resource vector, adverse benchmark, Path B contract, and falsification program. Evidence - not the direction of the initial intuition - must decide which restricted models, if any, support a speedup or a lower bound.

13. Genealogy and source provenance

The research program developed in stages. Dates below are dates of the named artifacts, not claims that every mature operator already existed at the first stage.

Date	Stage	Contribution preserved here	Provenance boundary
2025-10-04	Jackie Blankenship comparison exchange	Broader SIT coherence, field, and computation context entering the nearby source family.	Private exported AI exchange; context donor, not a P/NP theorem.
2025-10-05	“SIT and P vs NP” exchange	Micah Blumberg proposed the strong physical thesis that nature can settle encoded problems by coherence / entropy dynamics.	Author prompt is the proposal; the same exchange records the critique that the thesis lacked a uniform model, resource bounds, and standard proof.

Date	Stage	Contribution preserved here	Provenance boundary
2025-10-07	Branched P-vs-NP explanation	Multi-level explanation, local-check / global-assembly analogy, CGC model, Path B, K0-K6, barriers, and benchmark program were developed.	AI-assisted formalization and correction; mature Path B language is not backdated before this exchange.
2025-10-10	Heat-kernel / SIT comparison branch	Holonomy, Poincaré/log-Sobolev, gap, and macro-structure comparison scaffolding.	Comparison scaffold only; no P/NP or mass-gap proof.
2026-06-05	CORE development copies	The two principal exchanges were preserved in local research files.	Local file creation date proves custody of the copies, not origin of ideas already dated by the exports.
2026-06-12	Cluster Draft 6	Safe problem statement, conditional Path B target, barrier matrix, K-ledger, and source routing.	Private manuscript stage.
2026-06-13	BC-054IS	Finite counterexample to one arbitrary-endpoint bridge.	Exact formal obstruction only.
2026-06-16	BC-055O	Narrow finite constructor / listed-mediator repair.	Exact formal result only; no complexity consequence.
2026-07-07	Zenodo record 21231142	Public fixation of the first deposited paper version.	Public provenance; scientific claims remain bounded by the deposited text.
2026-07-11	Private v2 Draft 4	Primary resource-accounting references and formal case study integrated.	Private revision; all operational obligations remained open.
2026-07-27	This final-preprint successor	Earned-vocabulary rewrite, 13 numbered equations, restored coherence-time action and conditional convergence relation, exact adverse benchmark, claim atlas, and release validation.	New synthesis; no backdating of new equations, benchmark results, or normalized atlas.

The first authorial intuition and the later discipline are both part of the record. The October 5 proposal was stronger than the evidence supported. It was not erased; it was transformed into a list of operations and theorem obligations that can be tested. That transformation is the paper’s central methodological advance.

14. Claim and boundary ledger

ID	Statement	Class	Disposition
C01	P versus NP remains open.	Established fact	Retained; external primary sources [1,2].
C02	Efficient checking does not by itself supply efficient finding.	Established framing	Retained; formalized by Equation (1).
C03	SIT offers a physical vocabulary for constrained mismatch and coordinated update.	Interpretive model	Retained as an author proposal.
C04	CGC may provide useful instance-difficulty diagnostics.	Testable conjecture	Open; requires out-of-sample benchmarks.

ID	Statement	Class	Disposition
C05	CGC-0 fails on the bounded implication-chain and fixed-3-CNF cases reported here.	Reproducible finite result	Supported by exhaustive code and tests.
C06	CGC-0 results establish a general lower bound.	Prohibited promotion	Not claimed.
C07	C_{SIT} is a candidate physical cost measure.	Definition / proposal	Restored; operational validation open.
C08	Equation (9) is a convergence theorem.	Prohibited promotion	It is a retained schematic target only.
C09	Path B would have a standard consequence if every stated uniformity, resource, certificate, and readout condition were proved.	Conditional theorem target	Retained with assumptions inline.
C10	Any Encode / Evolve / Certify / Read / Check obligation is complete.	Prohibited promotion	All remain open.
C11	Any K0-K6 obligation is complete.	Prohibited promotion	All remain open.
C12	BC-054IS refutes one finite bridge.	Formal finite result	Replayed at exact scope.
C13	BC-055O proves a different narrow finite repair.	Formal finite result	Replayed at exact scope.
C14	The Branch-C artifacts imply P equals NP, P differs from NP, RH, HSIT, or full SIT.	Prohibited promotion	No such implication.
C15	A physical runtime is a Turing runtime without setup, precision, volume, probability, and readout accounting.	Rejected assumption	Replaced by Equation (7).
C16	$C(I) = gAM$ is a derived universal invariant.	Prohibited promotion	Retained only as a cross-domain measurement proposal.
C17	Equation (13) separates P from NP.	Prohibited promotion	It is a model-relative finite counting bound.
C18	The finite-napkin constraint or entropic-tunneling branch is proved.	Prohibited promotion	Both remain preserved historical hypotheses.

15. Conclusion

The intuition behind this paper is simple: checking a supplied pattern and assembling that pattern are physically different operations. SIT proposes that the assembly process may be understood through mismatch, coordinated differentials, coherence, dissipation, and macroscopic readout. The paper now earns those terms by first stating the ordinary problem, the established computational accounts, and the plain-language operation they are meant to name.

The bounded benchmark supplies the first adverse calibration. Lowering clause mismatch works on a friendly formula and fails in small, exactly enumerable landscapes. That failure does not refute the larger proposal. It identifies the missing content: collective motion, uniform construction, finite precision, macro closure, certificate production, robust readout, and complete resource accounting.

Path B is the strongest present formulation. If its encoder, dynamics or certificate constructor, margins, macro closure, readout, and ordinary verification can all be proved with polynomial resources on arbitrary NP-

complete instances, the consequence would be conventional. Until then, it is a theorem target and experiment program, not a Millennium Prize solution.

The scientific boundary is therefore exact:

Coherence-guided computation is a falsifiable program for organizing and measuring physical witness search. This paper supplies a model scaffold, adverse finite evidence, and a complete obligation ledger. It supplies neither a proof that P equals NP nor a proof that P differs from NP .

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Data, code, and formal-artifact availability

The release bundle accompanying this preprint contains the manuscript source, searchable PDF, deterministic CGC-0 benchmark and tests, normal and optimized benchmark reports, paper-local claim / source / genealogy

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Competing interests

The author declares no competing interests.